WORKFORCEAI

AI-Powered HR Workforce Insights Platform

---

1. Introduction

WorkforceAI is an AI-powered HR workforce insights platform designed to help HR teams monitor employee information, understand workforce patterns, and make data-informed decisions.

The application consolidates employee records, attendance, leave, attrition, onboarding, and workforce movement information into a centralized dashboard. It combines interactive data visualizations, machine learning-based predictions, statistical analysis, and a Generative AI chatbot.

The platform provides two interfaces: an HR Admin dashboard for organization-wide workforce analysis and an Employee Portal for accessing personal HR information.

2. Problem Statement

HR departments manage different types of employee information across multiple functions. Analyzing these records separately can make workforce monitoring and decision-making time-consuming.

The major challenges include:

- Difficulty obtaining a consolidated view of workforce information.
- Limited visibility into department-wise attrition and attendance patterns.
- Difficulty identifying employees who may be at risk of leaving.
- Challenges in identifying unusual attendance patterns.
- Limited ability to estimate future employee exits.
- Time-consuming retrieval of HR information through manual reports and dashboards.

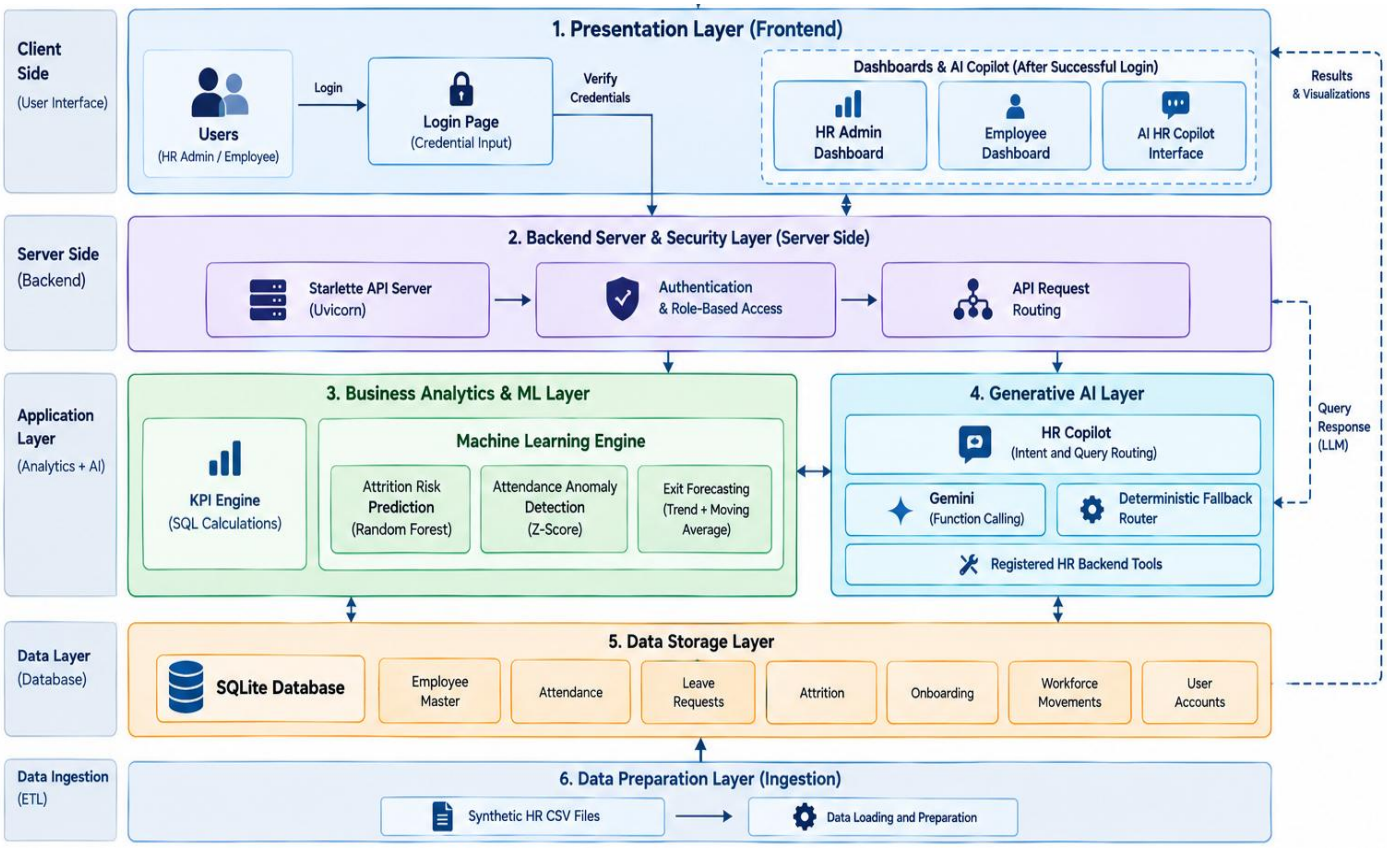
The objective is to develop a centralized HR workforce intelligence platform that combines data analytics, machine learning, and Generative AI to improve workforce visibility and support HR decision-making.

3. Proposed Solution

WorkforceAI integrates the following components:

- **Data Layer:** Stores employee and HR activity information in SQLite.
- **Backend:** Starlette-based APIs handle requests, database operations, and analytics.
- **Analytics Engine:** Calculates workforce KPIs and department-level insights.
- **ML Engine:** Provides attrition risk classification, attendance anomaly detection, and exit forecasting.
- **GenAI Chatbot:** Allows users to retrieve HR insights through natural-language questions.
- **Frontend:** Provides interactive dashboards and an employee self-service portal.

System Architecture



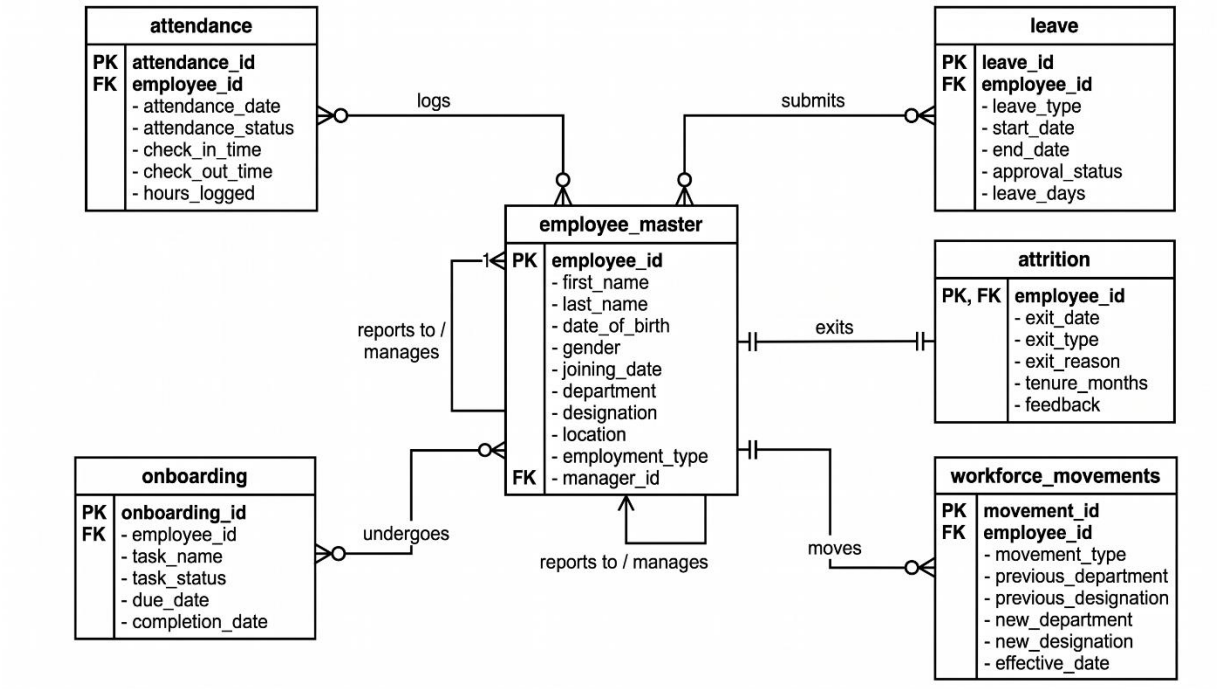
4. Dataset Description

The application uses synthetic HR data representing employee and organizational activities.

Dataset	Records	Purpose
Employee Master	500	Employee profile and employment details
Attendance	190,259	Employee attendance records
Leave	1,874	Leave requests and approval status
Attrition	70	Employee exit information
Onboarding	2,500	Employee onboarding tasks
Workforce Movements	59	Promotions, transfers and role changes

The six core datasets contain **195,262 records**.

Dataset ER Diagram



The Employee Master table acts as the central entity. Other tables reference employee_id to maintain relationships between employees and their HR activities.

5. Machine Learning and Predictive Analytics

WorkforceAI includes three analytical services.

5.1 Attrition Risk Prediction – Random Forest

Identify active employees who may have a higher likelihood of leaving the organization.

Random Forest is a supervised machine learning algorithm that combines multiple decision trees to predict an outcome.

Target variable: is_attrition (0 = Active, 1 = Exited).

Training configuration:

Parameter	Value
Algorithm	Random Forest Classifier
Number of trees	100
Maximum depth	5
Class weight	Balanced
Train-test split	80:20

Current evaluation results:

Metric	Result
Precision	0.375
Recall	0.214
F1-score	0.273
ROC-AUC	0.667

The model generates attrition risk scores. The dashboard groups active employees into High Risk (score ≥ 0.60) and Elevated Risk (score ≥ 0.40 and below 0.60).

5.2 Attendance Anomaly Detection – Z-Score

The Z-score method identifies attendance patterns that differ significantly from normal patterns.

$$Z = (x - \mu) / \sigma$$

This is statistical anomaly detection, not a trained ML model. An anomaly flag indicates a pattern that may require investigation.

5.3 Employee Exit Forecasting

The system estimates future employee exits using:

- Ordinary Least Squares (OLS) linear trend analysis.
- Three-month moving average.

Current forecast:

Month	Predicted Exits
November 2026	5
December 2026	5
January 2027	5
Total	**15**

This is a mathematical trend-based forecast, not a trained forecasting model.

6. GenAI Integration – HR Chatbot

The HR chatbot enables users to ask workforce-related questions in natural language.

Technology: Google Gemini API (gemini-3-flash-preview), integrated through Python REST API requests.

The HR Admin Copilot has nine registered analytics tools covering executive KPIs, department breakdown, attrition, attendance, leave, onboarding and mobility, flight risk, attendance anomalies, and exit forecasting.

The implementation supports two execution paths:

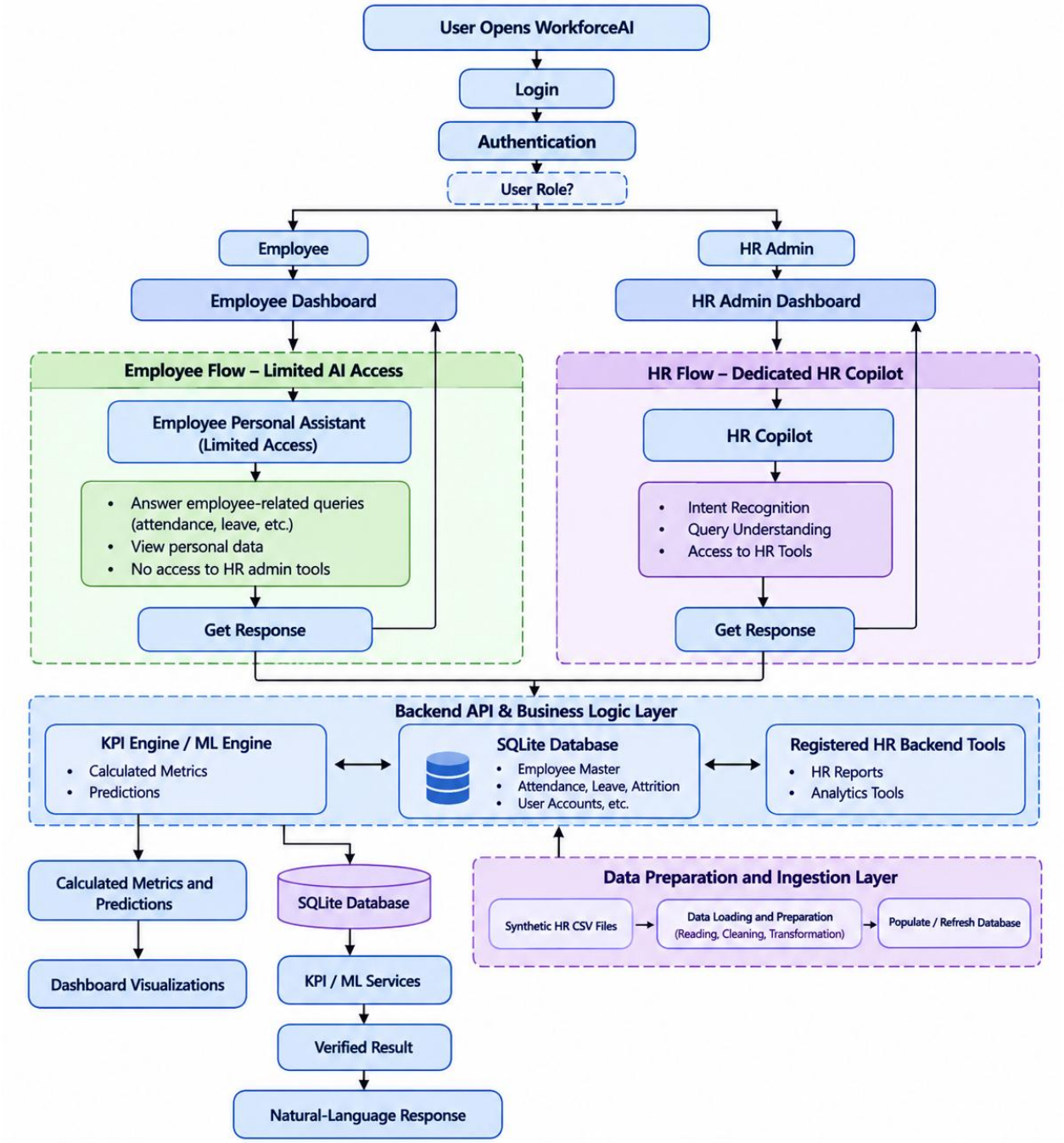
- **Gemini Function Calling:** Gemini selects an available backend function, the backend executes it, and the result is returned to Gemini to generate a response.

- **Deterministic Fallback:** If the Gemini path is unavailable, predefined intent and entity patterns route the query to an appropriate analytics function.

An Employee Copilot is also available with employee-specific information rather than organization-wide HR analytics tools.

7. System Functional Flow

The system follows this overall workflow:



8. Limitations

Current limitations:

- Attrition model recall is low and requires further improvement.
- Model validation requires stronger temporal separation and leakage prevention.
- ML services read CSV data separately from the SQLite database.
- Server-side authentication and role-based authorization require strengthening.
- Some leave calculations and business rules require correction.
- Live Gemini integration and complete browser-based testing require further verification..

9. Conclusion

WorkforceAI demonstrates the integration of HR data management, interactive analytics, machine learning, statistical analysis, and Generative AI into a centralized workforce intelligence platform.

It provides HR administrators with organization-wide workforce insights and employees with a personal self-service portal. The attrition classification model, attendance anomaly detection, exit forecasting, and Gemini-based chatbot demonstrate how AI and analytics can support HR decision-making.

The current application provides a functional prototype and a foundation for further improvements in predictive accuracy, security, and production readiness.